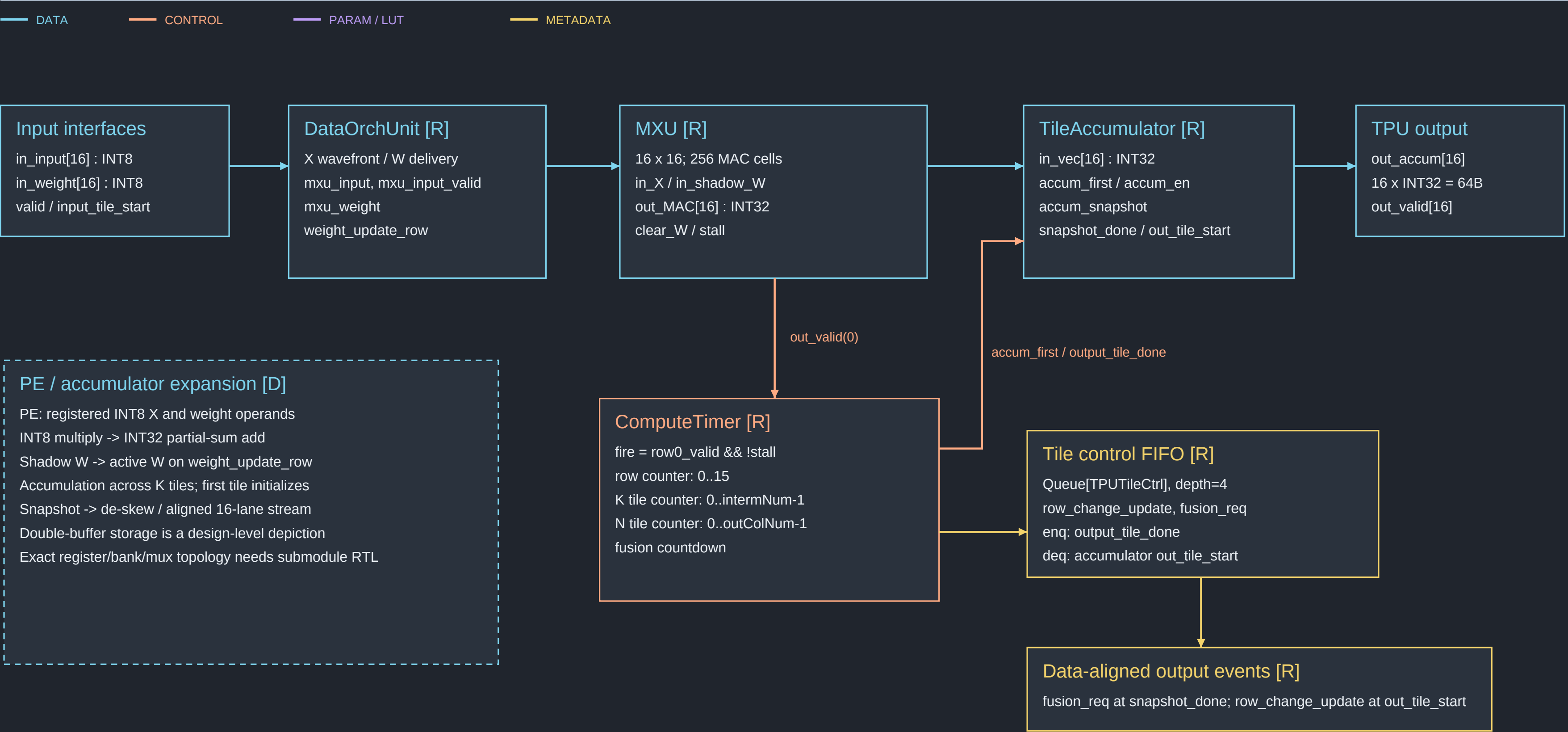


TPU / internal structure

R = connections inspected in saved TPU.scala and ComputeTimer.scala. Dashed detail is an architectural expansion without submodule RTL.

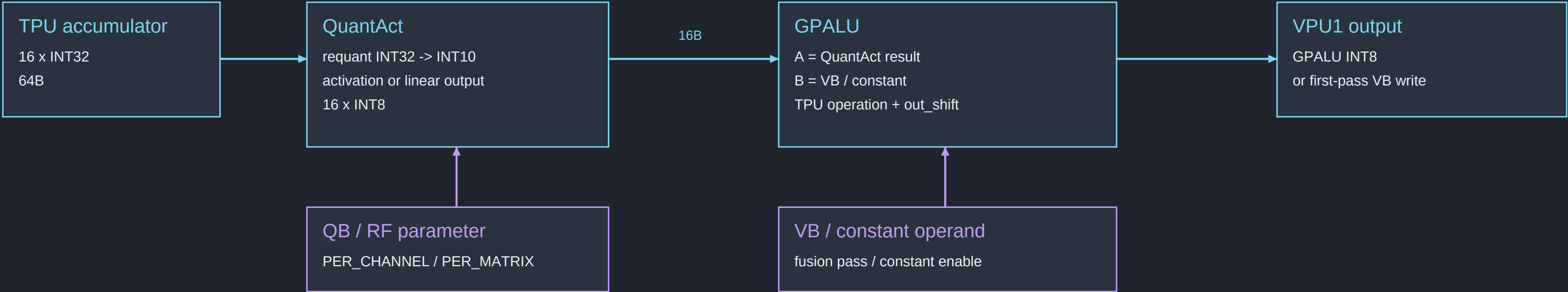


VPU1 / mode-dependent routing

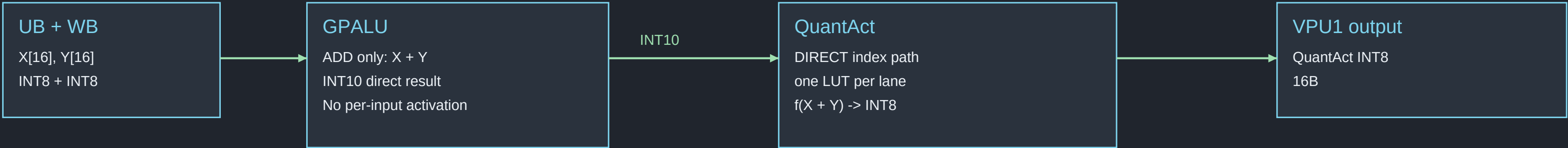
Frozen target: one QuantAct / activation LUT per lane and one GPALU datapath; the two routes below are mutually exclusive modes.

DATA CONTROL PARAM / LUT METADATA

TPU MODE



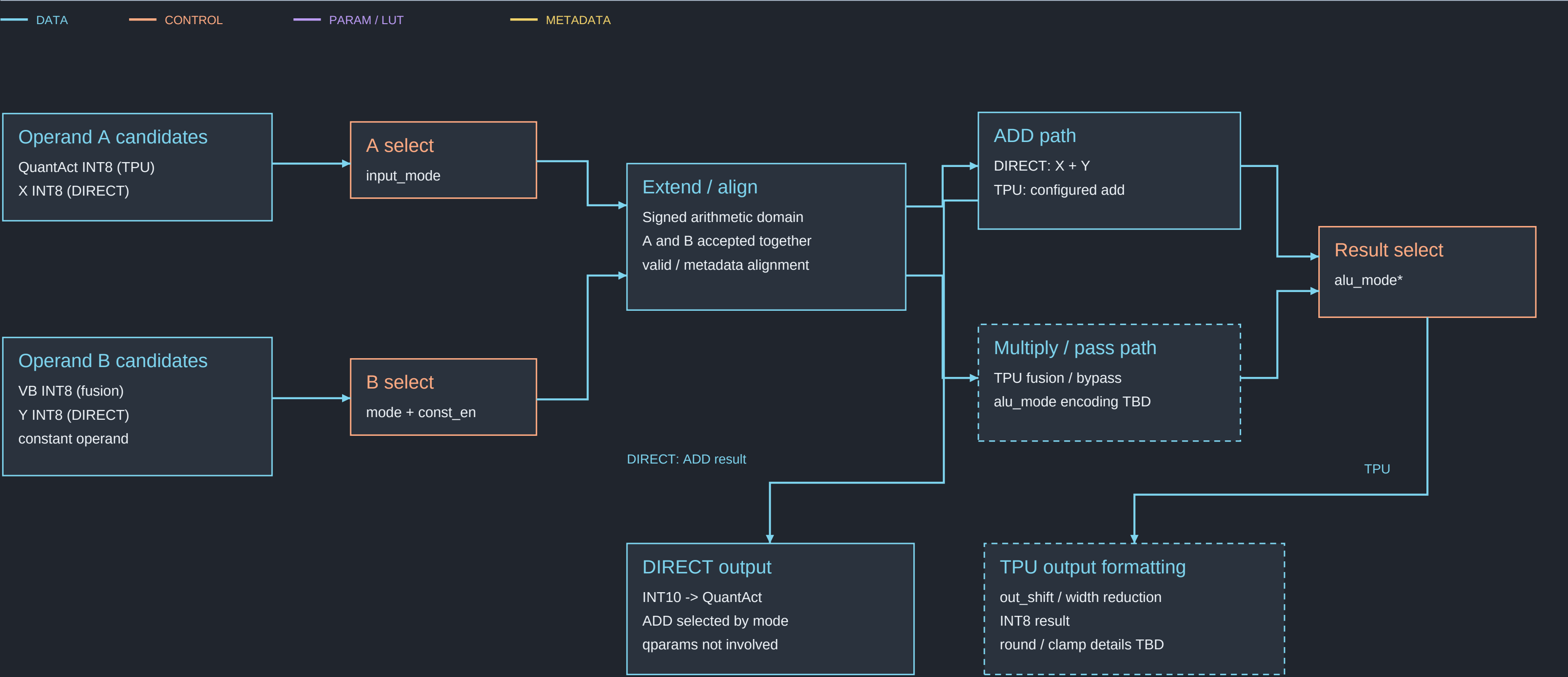
DIRECT MODE



Derived selection: qact_src_sel*=TPU/ALU; alu_a_sel*=QuantAct/X; vpu1_out_sel*=ALU/QuantAct.
DIRECT ignores qparams and suppresses QB requests. The two drawings do not imply duplicated hardware.
Controls: input_mode, act_mask, fusion_second, alu_mode, out_shift, out_lookahead, busy (handoff target names).
Source: CODEX_HANDOFF_QB64_FB32.md section 5E; 2026-09-09 user decision. Current VPU1 RTL not inspected.

GPALU / lane datapath

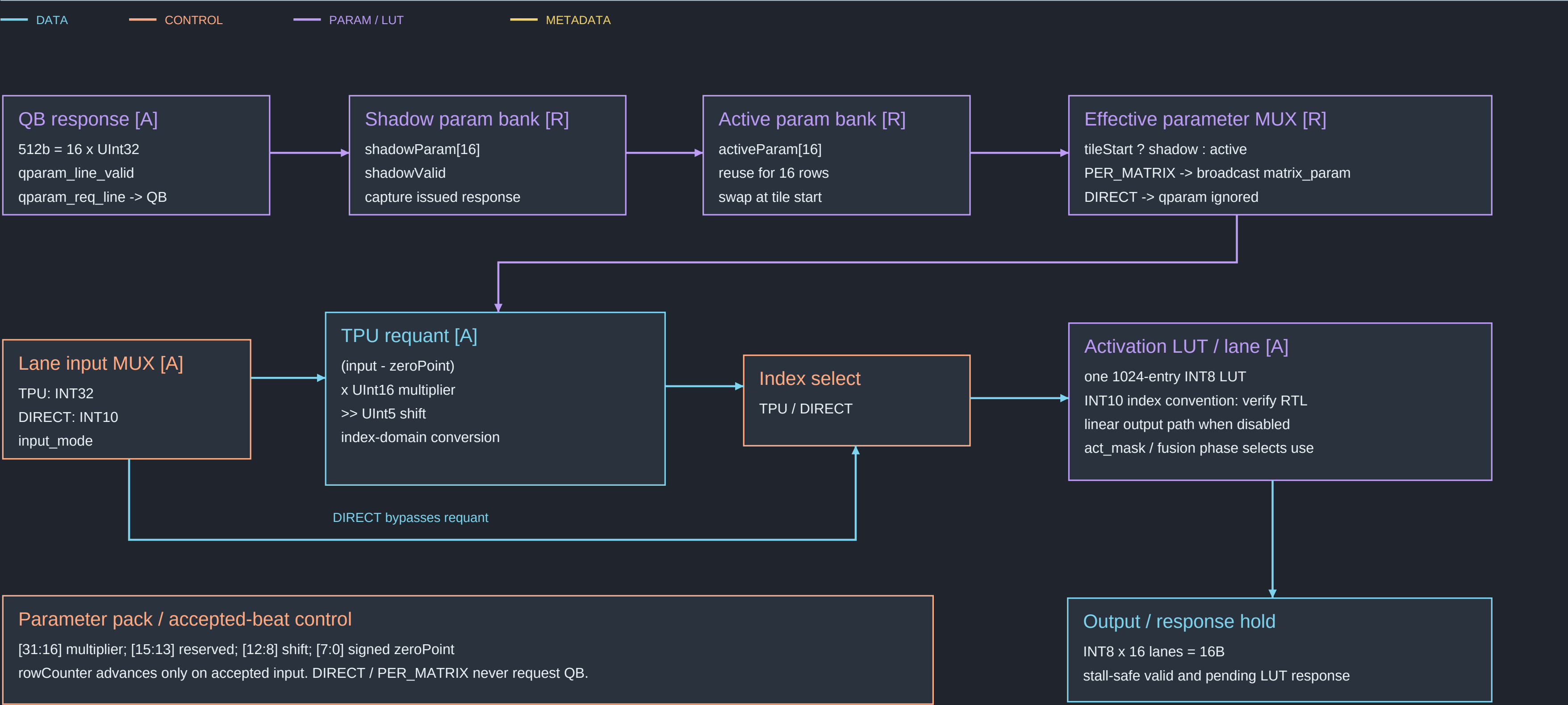
Architectural expansion [D]. Final mode behavior is known; register placement, enum values and exact scaling require current GPALU.scala.



This view repeats for 16 lanes. The operand muxes are logical selections, not verified RTL cell counts.
A signed INT8 sum needs 9 value bits; the agreed direct interface carries INT10 for QuantAct.
Only f(X+Y) is enabled for DIRECT. Separate f(X) and f(Y) LUT paths are excluded.
Source: frozen VPU1 route in the 2026-09-09 handoff and user decision; GPALU source was unavailable.

QuantAct / parameter engine & lane structure

Target [A] plus saved parameter-engine source [R]. Arithmetic rounding/index convention of the newer RTL remains to be checked.



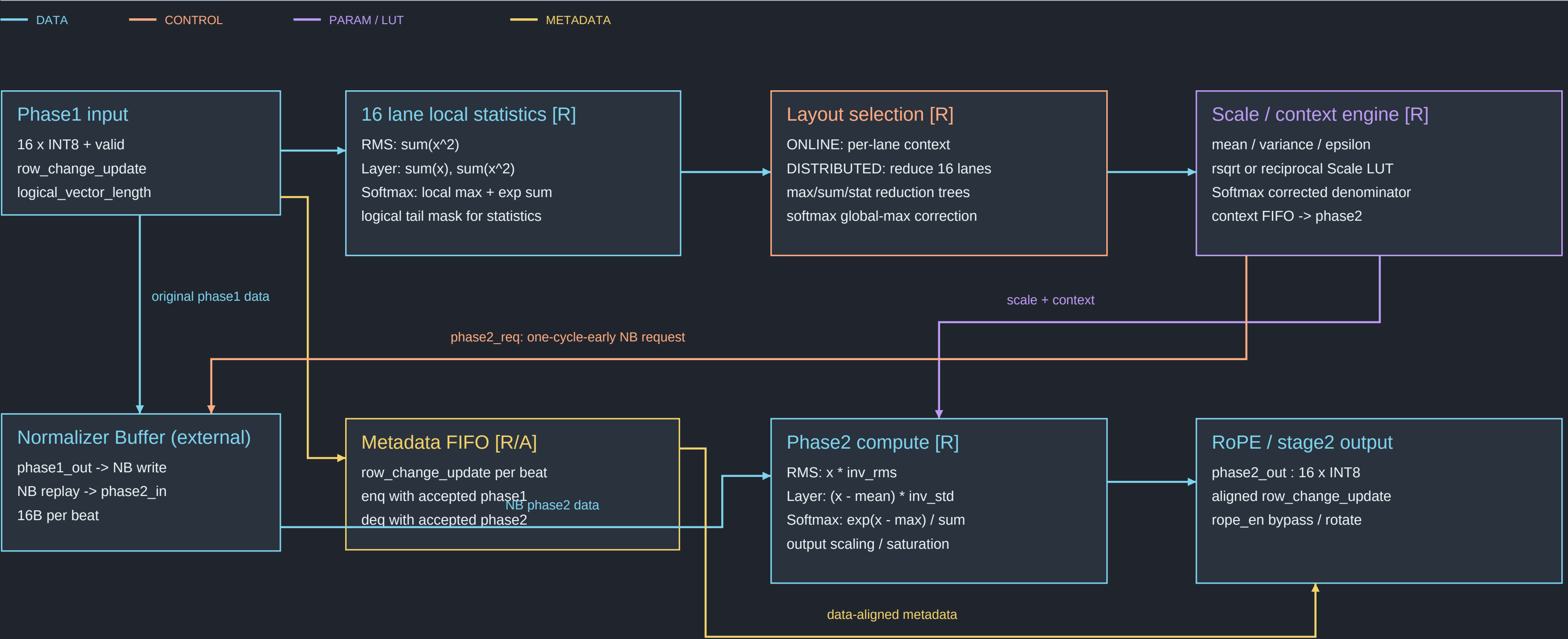
Saved QuantActUnit.scala predates the new DIRECT/fusion route. Its legacy shift-2/index clamp is not assumed to be final.

The diagram preserves the newer handoff requant formula without inventing the current INT10 indexing / rounding implementation.

Source: 02_QuantAct_QB_Spec.md; saved QuantActUnit.scala; CODEX_HANDOFF_QB64_FB32.md.

VPU2 / Normalizer internal structure

R = saved NormUnit_diststallfix.scala; row_change_update naming and 128b LUT programming follow the later target contract.



Mode encoding in saved source: 0 bypass, 1 RMSNorm, 2 LayerNorm, 3 Softmax. Layout is a separate configuration.

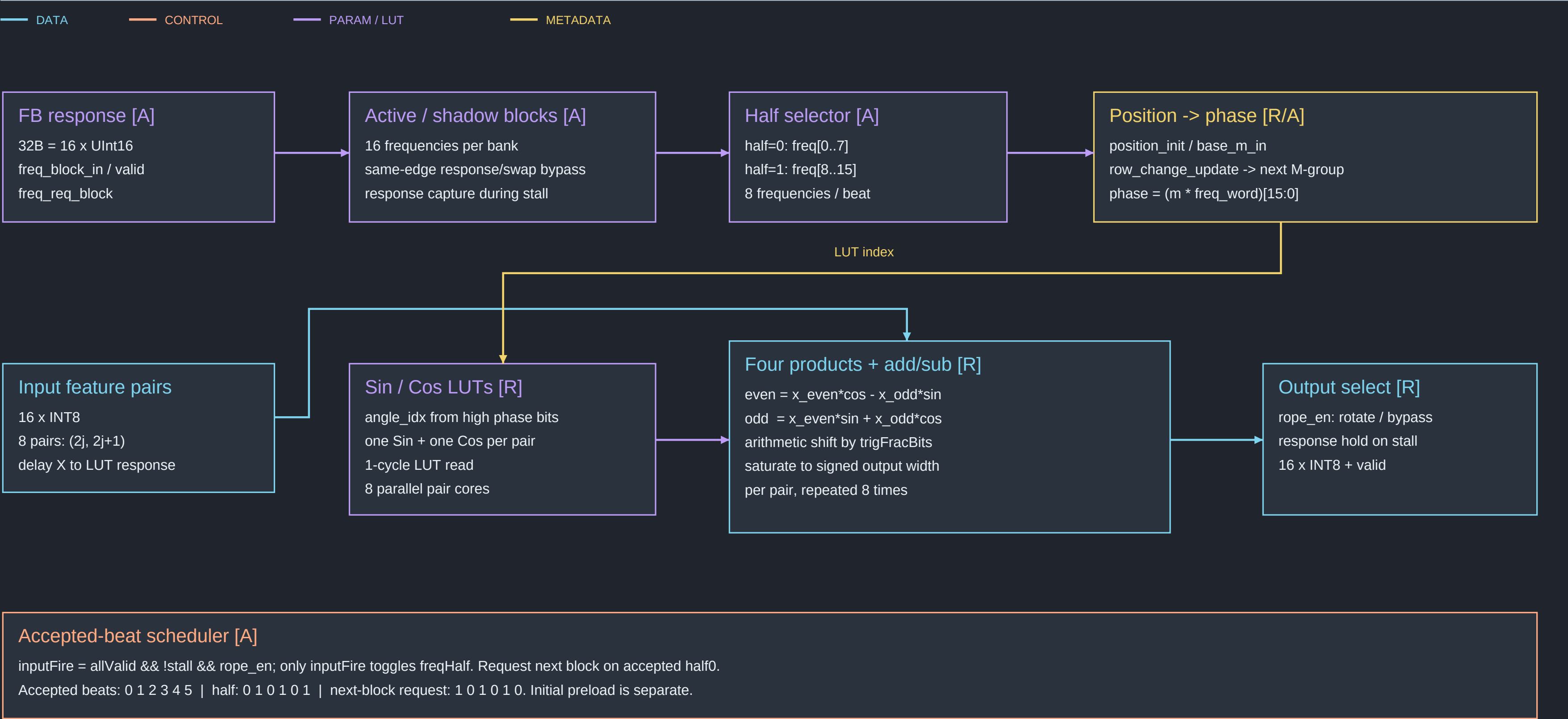
ONLINE uses row-change timing for vector boundary; DISTRIBUTED counts $\text{ceil}(\text{logical_length}/16)$ accepted beats.

Padding is excluded from phase1 statistics. Phase2 still processes physical beats; do not force LayerNorm padding outputs to zero.

Source: NormUnit_diststallfix.scala. Old phase*_rope_param_update_* ports map to target phase*_row_change_update_* names.

RoPE / frequency delivery and pair arithmetic

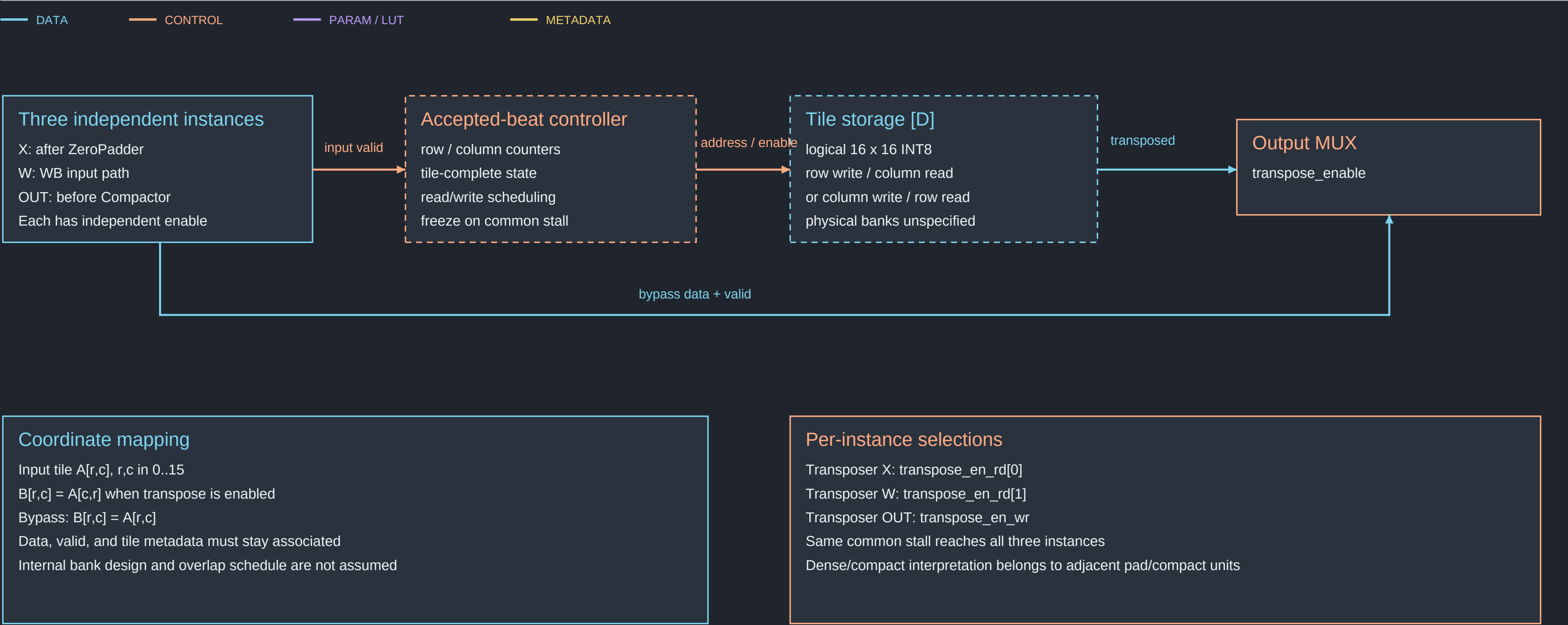
A = 32B / two-accepted-beat target; pair arithmetic and response hold inspected in saved Rope.scala [R].



Position changes and frequency fetch cadence are independent. No PB position reads; row_change_update stays aligned with data.
Saved Rope.scala still has the superseded 64B/four-group fetch. The 32B engine above is the later handoff target, not compiled evidence.
Source: Rope.scala (pair arithmetic); CODEX_HANDOFF_QB64_FB32.md sections 2-3 (frequency contract).

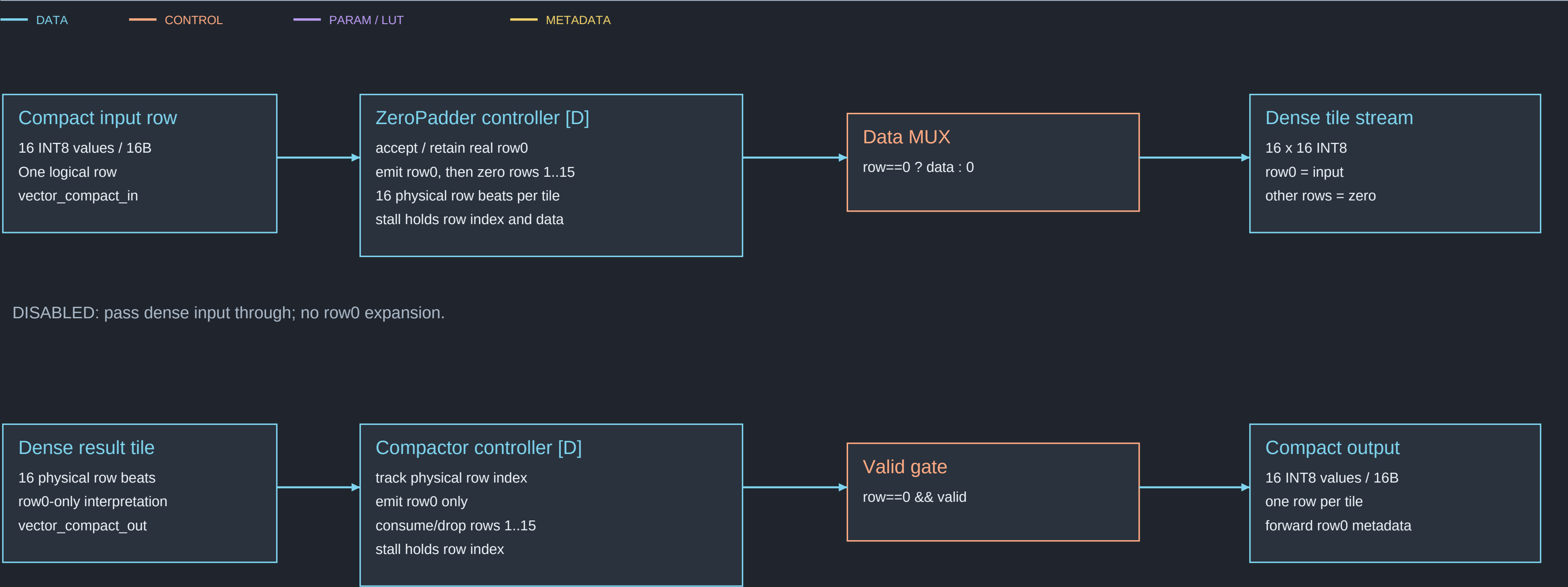
Three transposers / internal functional structure

Architectural expansion [D]. Instance placement is confirmed; exact RAM banking, port count, and register layout require transposer RTL.



ZeroPadder & Compactor / row0-only adapters

Frozen row0-only contract [A]; controller and mux internals are a functional decomposition [D].



LUT Programming Controller / selection and sequencing

Frozen 16B accepted-write contract [A]. Target/address counters below are a functional decomposition [D].

